
TE$_{01}$
TM$_{01}$
HE$^e_{21}$
HE$^o_{21}$
LP$_{11}$, $\hat{x}$ along lobes
LP$_{11}$, $\hat{x}$ across lobes
$E_t \propto (-y, x)$
$E_t \propto (x, y)$
$E_t \propto (x, -y)$
$E_t \propto (y, x)$
$= \frac{1}{2}(\mathrm{TM}_{01}+\mathrm{HE}^e_{21})$
$= \frac{1}{2}(\mathrm{HE}^o_{21}-\mathrm{TE}_{01})$
$E_z \neq 0$ on the axis ($\nabla \cdot E_t \neq 0$)
node in all components